Let $\{a_n\}_{n\geq 1}$ be a sequence (convergent or not).

<u>Case 1:</u> $\{a_n\}_{n\geq 1}$ is bounded.

Last time:

$u_N = \inf\{a_n : n \geq N\}$, $v_N = \sup\{a_n : n \geq N\}$ exist $\forall N \in \mathbb{N}$

Claim: $\{u_N\}_{N\geq 1}$ is increasing and $\{v_N\}_{N\geq 1}$ is decreasing. For $N \geq 1$,

$$\{a_n : n \geq N\} \supseteq \{a_n : n \geq N+1\} \Rightarrow \begin{cases} \inf\{a_n : n\geq N\} \leq \inf\{a_n : n \geq N+1\} \\ \sup\{a_n : n \geq N\} \geq \sup\{a_n : n \geq N+1\} \end{cases}$$

$$\Rightarrow \begin{cases} u_N \leq u_{N+1} \\ v_N \geq v_{N+1} \end{cases}$$

This yields

$$\star \quad u_1 \leq u_2 \leq \ldots \leq u_N \leq v_N \leq \ldots \leq v_2 \leq v_1 \qquad \forall N \geq 1$$

So $\{u_N\}_{N\geq 1}$ and $\{v_N\}_{N\geq 1}$ are both monotone and bounded, and therefore converge. Set

$$u = \lim_{N\to\infty} u_N \quad \text{and} \quad v = \lim_{N\to\infty} v_N .$$

By $\star$, we then have

$$u_M \leq v_N \quad \forall M, N \geq 1 \quad \Rightarrow \quad u \leq v_N \quad \forall N \geq 1$$

HW4#3

$$\Rightarrow \quad \boxed{u \leq v}$$

Moreover, if $\lim_{n\to\infty} a_n = a$ exists, then

$$u_N = \inf\{a_n : n \geq N\} \leq a_n \leq \sup\{a_n : n \geq N\} = v_N \quad \forall n \geq N$$

$$\underset{\text{HW4\#3}}{\Rightarrow} \quad u_N \leq a \leq v_N \qquad \underset{\text{HW4\#3}}{\Rightarrow} \quad \boxed{u \leq a \leq v}$$

<u>Def</u> Let $\{a_n\}_{n\geq 1}$ be a bounded sequence. We define

$$\limsup_{n\to\infty} a_n = \lim_{N\to\infty} v_N = \lim_{N\to\infty} \left(\sup\{a_n : n \geq N\}\right)$$

$$\liminf_{n\to\infty} a_n = \lim_{N\to\infty} u_N = \lim_{N\to\infty} \left(\inf\{a_n : n \geq N\}\right)$$

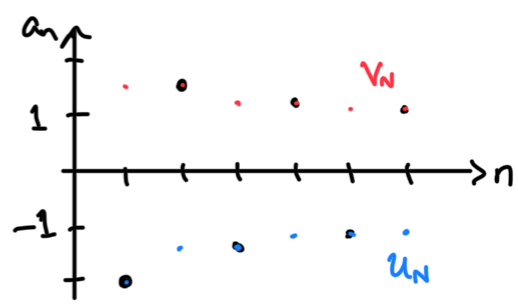
- These make sense even when $\{a_n\}_{n\geq 1}$ diverges
- The first boxed inequality proves:

<u>Cor</u> If $\{a_n\}_{n\geq 1}$ is bounded, then

$$\liminf_{n\to\infty} a_n \leq \limsup_{n\to\infty} a_n .$$

<u>Ex</u> $\{a_n\}_{n\geq 1}$ where $a_n = (-1)^n\left(1+\frac{1}{n}\right)$

- $\limsup_{n\to\infty} a_n = 1$
- $\liminf_{n\to\infty} a_n = -1$
- $\lim_{n\to\infty} a_n$ does not exist



<u>Case 2:</u> $\{a_n\}_{n\geq 1}$ is unbounded.

① Not bounded above:

- $v_N = \sup\{a_n : n \geq N\}$ does not exist $\forall N \geq 1$
- if $\{u_N\}_{N\geq 1}$ increasing and $\begin{cases} \text{bounded} \Rightarrow \lim u_N \in \mathbb{R} \\ \text{unbounded} \Rightarrow \lim u_N = +\infty \end{cases}$

② Not bounded below:

- $u_N = \inf\{a_n : n \geq N\}$ does not exist $\forall N \geq 1$
- if $\{v_N\}_{N\geq 1}$ decreasing and $\begin{cases} \text{bounded} \Rightarrow \lim v_N \in \mathbb{R} \\ \text{unbounded} \Rightarrow \lim v_N = -\infty \end{cases}$

<u>Def</u> ① If $\{a_n\}_{n\geq 1}$ is not bounded above, then we define

$$\limsup_{n\to\infty} a_n = +\infty.$$

② If $\{a_n\}_{n\geq 1}$ is not bounded below, then we define

$$\limsup_{n\to\infty} a_n = -\infty.$$

- So limsup and liminf make sense for any sequence, but their values are in $\mathbb{R} \cup \{\pm\infty\}$

<u>Cor</u> For any sequence $\{a_n\}_{n\geq 1}$, we have

$$\liminf_{n\to\infty} a_n \leq \limsup_{n\to\infty} a_n.$$

- The only new cases are $\liminf_{n\to\infty} a_n = \pm\infty$ or $\limsup_{n\to\infty} a_n = \pm\infty$. Of course, $-\infty < x < \infty$ $\forall x \in \mathbb{R}$.

<u>Thm</u> Let $\{a_n\}_{n\geq 1}$ be a sequence.

① If $\lim_{n\to\infty} a_n$ exists in $\mathbb{R} \cup \{\pm\infty\}$, then

$$\liminf_{n\to\infty} a_n = \lim_{n\to\infty} a_n = \limsup_{n\to\infty} a_n.$$

② If $\liminf_{n\to\infty} a_n = \limsup_{n\to\infty} a_n$ in $\mathbb{R} \cup \{\pm\infty\}$, then

$$\lim_{n\to\infty} a_n = \liminf_{n\to\infty} a_n = \limsup_{n\to\infty} a_n .$$

- Here, "$\lim_{n\to\infty} a_n$ exists in $\mathbb{R}\cup\{\pm\infty\}$" just means "$\{a_n\}_{n\geq 1}$ converges, or diverges to $+\infty$, or diverges to $-\infty$".

<u>Pf</u>: ① There are 3 cases.

Case 1: $\lim_{n\to\infty} a_n = -\infty$. It suffices to show $\limsup_{n\to\infty} a_n = -\infty$, since $\liminf_{n\to\infty} a_n \leq \limsup_{n\to\infty} a_n$.

Fix $M<0$. As $\lim_{n\to\infty} a_n = -\infty$, $\exists\, N_0 \in \mathbb{N}$ s.t.

$$a_n < M-1 \quad \forall n \geq N_0$$

$$\Rightarrow \quad v_{N_0} = \sup\{a_n : n \geq N_0\} \leq M-1$$

$$\Rightarrow \quad v_N \underset{\substack{\uparrow \\ v_N \text{ decreasing}}}{\leq} v_{N_0} \leq M-1 < M \quad \forall N \geq N_0$$

So $\lim_{N\to\infty} v_N = -\infty$, and therefore $\limsup_{n\to\infty} a_n = -\infty$ as desired.

Case 2: $\lim_{n\to\infty} a_n = +\infty$. Similar. (Exercise.)

Case 3: $\lim_{n\to\infty} a_n = a \in \mathbb{R}$. Fix $\varepsilon > 0$. Then $\exists\, N_0 \in \mathbb{N}$ s.t.

$$|a_n - a| < \frac{\varepsilon}{2} \quad \forall n \geq N_0$$

$$\Rightarrow \quad a - \frac{\varepsilon}{2} < a_n < a + \frac{\varepsilon}{2} \quad \forall n \geq N_0$$

$$\Rightarrow \quad a - \frac{\varepsilon}{2} \leq \inf\{a_n : n \geq N_0\} \leq \sup\{a_n : n \geq N_0\} \leq a + \frac{\varepsilon}{2}$$

$$\Rightarrow \quad a - \frac{\varepsilon}{2} \le u_{N_0} \le u_{N_0+1} \le \dots \le u \le v \le \dots \le v_{N_0+1} \le v_N \le a + \frac{\varepsilon}{2}$$

$$\Rightarrow \quad |u_N - a| \le \frac{\varepsilon}{2} < \varepsilon, \qquad |v_N - a| \le \frac{\varepsilon}{2} < \varepsilon \quad \forall N \ge N_0$$

So $\liminf\limits_{n\to\infty} a_n = \lim\limits_{N\to\infty} u_N = a$ and $\limsup\limits_{n\to\infty} a_n = \lim\limits_{N\to\infty} v_N = a$.

② Next time!